

glucose to survive. The compounds present in any given cell at a certain moment determine which operons are switched on—the result of simple interactions of activator and repressor proteins with the promoters of the genes in question.

CONCEPT CHECK 18.1

1. How does binding of the *trp* corepressor to the *trp* repressor alter repressor function and transcription? How does binding of the *lac* inducer alter the function of the *lac* repressor?
2. Describe the binding of repressors and activators to the *lac* operon when both lactose and glucose are scarce. What is the effect of these scarcities on transcription of the *lac* operon?
3. **WHAT IF?** A certain mutation in *E. coli* changes the *lac* operator so that the active repressor cannot bind. How would this affect the cell's production of β -galactosidase?

For suggested answers, see Appendix A.

CONCEPT 18.2

Eukaryotic gene expression is regulated at many stages

All organisms, whether prokaryotes or eukaryotes, must regulate which genes are expressed at any given time. Both unicellular organisms and the cells of multicellular organisms continually turn genes on and off in response to signals from their external and internal environments. Regulation of gene expression is also essential for cell specialization in multicellular organisms, which are made up of different types of cells. To perform its own distinct role, each cell type must maintain a specific program of gene expression in which certain genes are expressed and others are not (see Figure 18.1).

Differential Gene Expression

A typical human cell might express about a third to a half of its protein-coding genes at any given time. Highly differentiated cells, such as muscle or nerve cells, express a smaller fraction of their genes. Almost all the cells in a multicellular organism contain an identical genome. (Cells of the immune system are one exception, as you will see in Figure 43.13.) A subset of genes is expressed in each cell type; some of these—about 35%—are “housekeeping” genes, expressed by many cell types, while others are unique to that cell type. The uniquely expressed genes allow these cells to carry out their specific function. The differences between cell types, therefore, are due not to different genes being present, but to **differential gene expression**, the expression of different genes by cells with the same genome.

The function of any cell depends on its expressing the appropriate set of genes. The transcription factors of a cell must locate the right genes at the right time, a task like finding a needle in a haystack. Abnormal gene expression can cause serious imbalances and diseases, including cancer.

Figure 18.6 summarizes the process of gene expression in a eukaryotic cell, highlighting key stages in the expression of a protein-coding gene. Each stage depicted in Figure 18.6

▼ **Figure 18.6 Stages in gene expression that can be regulated in eukaryotic cells.** In this diagram, the colored boxes indicate the processes most often regulated; each color indicates the type of molecule that is affected (blue = DNA, red/orange = RNA, purple = protein). The nuclear envelope separating transcription from translation in eukaryotic cells allows for RNA processing, a form of post-transcriptional control that is not possible in prokaryotes. Eukaryotes also have a greater variety of control mechanisms operating before transcription and after translation. Some genes are regulated at multiple stages.

